■ Neural Sound iOS — Executive Daily Analytics & Growth Intelligence

■ Product Overview & Health Score

Overall Health Score: 7.8 / 10

Mathematical Rationale:

- **Acquisition Momentum (+3.0 / 3.0):** Phenomenal DAU expansion (+342% over 7 days, rising from 33 to 146 daily active users), heavily powered by new user volume (+2,575%).
- **Engagement & Session Depth (+2.5 / 2.5):** Industry-leading engagement rate (**84.9%**) and average session duration (**10.4 minutes** across all users; **18.7 minutes** for returning power users).
- **Funnel Health & Onboarding (+1.8 / 2.0):** Flawless top-of-funnel authentication/onboarding completion (**97.2%** conversion from `first_open` to `first_journey_login_completed`).
- **Telemetry & Release Stability (-1.5 / 2.5):** Critical telemetry flaws (un-debounced audio sliders generating 5,700+ events/day across just 10 users) and a alarming engagement drop in staged version `1.3.08` (**53.8%** vs **86.8%** in `1.3.07`).

Executive Summary:

- **Surging Volume:** Neural Sound iOS achieved **146 Active Users** and **186 Sessions** yesterday, experiencing a massive traffic surge driven primarily by viral organic acquisition in South Korea (**75.3% of total DAU**).
- **Power User Stickiness:** While new users average **5.8 minutes** per session, returning power users spend **18.7 minutes** per session, demonstrating deep utility in audio stem separation, pitch, and tempo manipulation.
- **Key Anomalies:**
- **Analytics Noise:** Continuous slider drag events (`player_pitch_changed`, `player_tempo_changed`) fire raw events without debouncing (up to **570 events/user**), polluting event payloads.
- **Aggressive Review Prompts:** `rate_prompt_shown` triggered **123 times across 122 users**, meaning almost every new user was prompted for a rating on Day 0.
- **Staged Rollout Alert:** Version `1.3.08` shows a severe drop in engagement rate (**53.8%**) compared to production release `1.3.07` (**86.8%**).

■ 1. Old vs. New User Cohort Deep-Dive

Cohort	Active Users	% of Active	Total Sessions	Sessions / User	Avg Session Duration	Engagement Rate	Screenviews / User
New Users	114	72.2%	112	0.98	347.8s (5.8 min)	**94.6%**	2.01 (229 total)
Returning Users	44	27.8%	69	1.57	**1,120.8s (18.7 min)**	**73.9%**	**3.18 (140 total)**
(not set)	0	0.0%	5	N/A	6.5s (0.1 min)	20.0%	0.02 (1 total)

Behavioral Analysis & Retention Drop-Off Risk

- **New User Dynamics:** New users show an exceptionally high initial engagement rate (**94.6%**) and spend nearly 6 minutes during their first onboarding and setup session. They quickly pass through account creation and experiment with stem separation.
- **Returning Power Users:** Returning users represent the core product-market fit. They execute longer sessions (**18.7 min**), view more screens per session (**3.18 screenviews**), and repeatedly manipulate audio stems, pitch, and tempo.
- **Retention Leakage Risk:** Despite top-of-funnel acquisition spiking by **+2,575%** (4 new users on Aug 15 vs 107 on Aug 21), returning active users only grew by **+34.5%** (29 returning users on Aug 15 vs 39 on Aug 21). **New users are engaging deeply on Day 0, but D1–D7 retention conversion remains a key bottleneck.**

■ 2. 7-Day Performance & Velocity Trajectory

Date	DAU	New Users	Returning Users	Sessions	Engagement Rate	Visual Velocity Trajectory
2026-08-15	33	4	29	57	78.9%	`[███████████] 33`
2026-08-16	30	7	23	47	80.9%	`[███████████] 30`
2026-08-17	37	14	23	48	81.2%	`[███████████] 37`
2026-08-18	59	37	22	78	83.3%	`[███████████] 59`
2026-08-19	102	84	18	137	86.9%	`[███████████] 102`
2026-08-20	94	65	29	116	81.9%	`[███████████] 94`
2026-08-21	**146**	**107**	**39**	**186**	**84.9%**	`[███████████] 146`

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7-Day DAU Growth Momentum: 33 Users ████ 30 ████ 37 ████ 59 ████ 102 ████ 94 ████ 146 Users (+342%
7-Day Velocity)
```

Momentum Analysis:

- **Acquisition Trajectory: Exponential Acceleration**. The surge beginning August 18 represents a breakout organic loop or external feature campaign.
- **Quality Preservation:** As DAU increased 4.4x, the overall **Engagement Rate remained high (84.9%)**, confirming that acquired users are high-intent target users rather than low-quality ad clicks.

■ 3. End-to-End User Journey & Funnel Diagrams

Diagram 1: Onboarding & Auth Funnel

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[ App First Open ] (109 Users / 109 Events) ■ ▼ 100.0% Step Conversion [ Onboarding Viewed ] (111 Users /
113 Events) ■ ▼ 99.1% Step Conversion [ Login Succeeded ] (110 Users / 110 Events) ■ ▼ 96.4% Step
Conversion [ First Journey Done ] (106 Users / 106 Events)
█████████████████████████████████████████████████████ OVERALL FUNNEL CONVERSION (First
Open ███ Journey Done): 97.2% Friction Risk: MINIMAL (Exceptional iOS Auth Flow)
```

Diagram 2: Core Audio Creation & Player Funnel

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[ Plus Button Tapped ] (105 Unique Users / 225 Events) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼ 74.3% Initiated Import ▼ 56.2% Initiated Import [
Import Files App Tapped ] (78 Users) [ Import Photos Tapped ] (63 Users) ■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼ [ Import Started ] (53 Unique Users / 105 Events)
■■ BOTTLENECK 1: 49.5% Drop-off from Plus Tap to Import Completion ■ ▼ 100.0% Transferred (Includes Ret
Users) [ Separation Player Opened ] (72 Unique Users / 199 Events) ■ ▼ 65.3% Conversion to Playback [
Separation Play Started ] (47 Unique Users / 571 Events) ■■ BOTTLENECK 2: 34.7% Drop-off from Player Opened
to Audio Playback ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ OVERALL CORE
FUNNEL CONVERSION (Plus Tap ■■■ Play Started): 44.8%
```

High-Friction Drop-Off Identification:

1. **File Selection & Processing Drop-off (49.5% Loss):** 105 users tap `plus_button_tap`, but only 53 successfully complete `import_started`. This indicates file format incompatibility, permission prompts blocking access, or large file processing latency before import confirmation.
2. **Player Execution Drop-off (34.7% Loss):** 72 users open `separation_player_opened`, but only 47 trigger `separation_play_started`. Users are landing on the player screen without audio automatically playing or encountering stem processing delay banners.

■■ 4. Feature Telemetry & Power-User Lookups

Event Name	Event Count	Total Users	Events / User	Feature Analysis & Telemetry Anomalies
`player_pitch_changed`	**5,706**	10	**570.6**	■ **CRITICAL BUG:** Continuous slider tracking un-debounced. Fires on every pixel change.
`player_tempo_changed`	**2,046**	9	**227.3**	■ **CRITICAL BUG:** Continuous slider tracking un-debounced. Fires on every pixel change.
`audio_sessions`	1,265	75	16.9	Core audio engine initialization event; healthy active utilization.
`user_engagement`	1,187	146	8.1	Native Firebase heartbeat event.
`separation_play_started`	571	47	12.1	Core playback execution; high repetition among active listeners.
`onboarding_step_viewed`	449	111	4.0	Indicates a 4-step onboarding slide sequence.
`screen_view`	370	112	3.3	General screen transitions.

`onboarding_next_tapped`	336	110	3.1	Navigation progression in onboarding.
`separation_page_viewed`	327	165	2.0	High surface visibility across new and returning users.
`plus_button_tap`	225	105	2.1	High intent entry point for creation.
`player_stem_muted`	209	5	**41.8**	Power feature usage; vocal/instrument isolation testing.
`separation_player_opened`	199	72	2.8	Primary audio utility interface navigation.
`player_stem_balance_changed`	132	1	**132.0**	Single power user deeply tweaking stem volumes (Un-debounced slider).
`rate_prompt_shown`	123	122	**1.0**	■■ **AGRESSIVE TRIGGER:** Fired for ~100% of new users regardless of value delivered.

■ 5. App Version Adoption & Global Footprint

App Version Distribution

App Version	Active Users	% Share	Total Sessions	Engagement Rate	Status & Stability Notes
1.3.07	136	93.2%	152	**86.8%**	**Production Main:** Highly stable, primary production build.
1.3.08	5	3.4%	13	**53.8%**	■■ **Staged Rollout Alert:** Severe engagement drop (**-33.0%** vs 1.3.07).
1.0.01	3	2.1%	17	88.2%	Legacy version; persistent long-tail power users.
1.3.06	2	1.4%	4	100.0%	Legacy build; near full deprecation.

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Version Adoption Visualization: v1.3.07 [██████████████████████████████████████████████████] 93.2% v1.3.08
[██████████████████████████████████████████████████] 3.4% (ENGAGEMENT DROP ALERT) v1.0.01
[██████████████████████████████████████████████████] 2.1% v1.3.06
[██████████████████████████████████████████████████] 1.4%
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Global Geographic Footprint

Country	Active Users	% DAU Share	Total Sessions	Sessions / User	Growth & Market Context
South Korea	**110**	**75.3%**	**126**	1.15	**Primary Viral Hub:** Dominates current traffic.
Bangladesh	8	5.5%	30	3.75	High frequency repeat sessions.
Iran	7	4.8%	9	1.29	Organic long-tail adoption.
India	4	2.7%	2	0.50	Emerging audience.
Japan	3	2.1%	3	1.00	High ARPU target region.
United States	3	2.1%	2	0.67	Key tier-1 expansion market.
Other (Canada, FR, PH, OM)	6	4.1%	8	1.33	Rest of World aggregation.

■ 6. Strategic Action Matrix (Prioritized)

Priority	Category	Issue / Finding	Recommended Action	Expected Impact
P0	**Engineering**	**Slider Telemetry Flooding:** `player_pitch_changed` and `player_tempo_changed` generate ~8,000 spam events daily due to continuous touch logging.	Replace drag events with `UIControl.Event.editingDidEnd` or implement a **500ms debounce buffer** on slider change listeners in Swift.	Reduces overall analytics payload volume by **~65%**, preventing Firebase rate limiting and billing inflation.
P0	**Engineering / Product**	**v1.3.08 Performance Degradation:** Staged rollout build `1.3.08` engagement rate dropped to **53.8%** (vs 86.8% in v1.3.07).	Immediately pause the 1.3.08 staged rollout in App Store Connect; review crash reports, audio processing engine logs, and UI thread blocking.	Protects new cohort experience and prevents negative App Store reviews.
P1	**Product / Growth**	**Premature Rating Prompts:** `rate_prompt_shown` triggers on 122/109 new users on Day 0 before core audio value is realized.	Restrict `SKStoreReviewController` triggering to post-event conditions: **after a user successfully completes 2+ stem separations**.	Prevents review fatigue, avoids store penalties, and increases 5-star rating conversion.

P1	**Growth**	**South Korea Acquisition Surge:** **75.3%** of DAU originated in South Korea, but app localization and Korean-specific keywords may be lacking.	Localize App Store metadata (Korean title, subtitle, keywords, screenshots) and launch targeted outreach to local music schools/K-Pop cover creators.	Capitalizes on viral momentum to convert viral traffic into a permanent growth engine.
P2	**Product**	**Creation Funnel Drop-off: 49.5%** drop-off between `plus_button_tap` and `import_started`; **34.7%** drop-off before starting playback.	Add audio file loading indicators, support drag-and-drop iOS file imports, and implement an auto-play toggle when opening the player.	Increases D0 core feature conversion from 44.8% to **>60%**, accelerating D1 retention.